

**Department of Computer Science & Engineering (IoT)****Vision of the Department***To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.***Mission of the Department***To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.***Session 2026-2027**

Vision: Dream of where you want.	Mission: Means to achieve Vision
---	---

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Sheikh Wasimuddin
Name of Student

**Department of Computer Science & Engineering (IoT)****Vision of the Department***To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.***Mission of the Department***To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.*

Session	2026-27 (ODD)	Course Name	Lab: Hybrid Computing Systems
Semester	7	Course Code	23IOT1704
Roll No	61	Name of Student	Sheikh Wasimuddin

Practical Number	01
Course Outcome	1. Understand Cloud Computing architecture, service models, deployment models, and virtualization. 2. Apply cloud platforms, storage services, and containerization for application deployment. 3. Implement Edge and Fog Computing solutions for IoT applications. 4. Analyze and develop integrated Cloud-Fog-Edge computing applications based on performance evaluation.
Aim	To study Cloud Computing Architecture, Service Models (IaaS, PaaS, SaaS), and Deployment Models (Public, Private, Hybrid, Community).
Theory (100 words)	Cloud Computing is a model in which computing services are delivered through the Internet instead of depending entirely on local computer systems. It provides flexible access to resources such as processing power, memory, storage, databases, and software. Cloud architecture generally consists of a client side, which is used by customers to access services, and a provider side, where computing resources are hosted and managed. Cloud services are commonly classified into IaaS, PaaS, and SaaS. IaaS offers virtualized infrastructure, PaaS provides an application development environment, and SaaS delivers ready-to-use applications. Cloud environments can be deployed as Public, Private, Hybrid, or Community clouds depending on ownership, accessibility, security, and organizational requirements. Service Models • IaaS (Infrastructure as a Service): Provides virtualized resources such as servers, storage, and networking. Customers have control over the operating system and applications while the provider manages the physical infrastructure.



Department of Computer Science & Engineering (IoT)

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.

	Example: Amazon EC2. • PaaS (Platform as a Service): Provides tools, runtime environments, databases, and platforms required to create and deploy applications. It reduces the need to manage servers and operating systems. Example: Google App Engine. • SaaS (Software as a Service): Provides fully developed applications through the Internet. Users can access the software using a browser without handling installation, maintenance, or infrastructure. Example: Gmail, Google Docs. Deployment Models • Public Cloud: Cloud infrastructure is provided by a third-party provider and is accessible to multiple customers through the Internet. • Private Cloud: Cloud resources are dedicated to one organization, providing greater control over infrastructure, access, and data management. • Hybrid Cloud: Combines private and public cloud environments, allowing data and applications to operate across both environments when required. • Community Cloud: Infrastructure is shared by organizations having common objectives, regulatory requirements, security needs, or operational policies.
Procedure and Execution (100 Words)	1. Understand the fundamental concept of Cloud Computing and its role in delivering computing resources over the Internet. 2. Examine the basic Cloud Computing architecture by identifying the Front End, Back End, and the Internet that enables communication between them. 3. Analyze the major Cloud Service Models: ○ IaaS – Infrastructure as a Service ○ PaaS – Platform as a Service ○ SaaS – Software as a Service 4. Examine the different Cloud Deployment Models: ○ Public Cloud ○ Private Cloud ○ Hybrid Cloud



Department of Computer Science & Engineering (IoT)

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

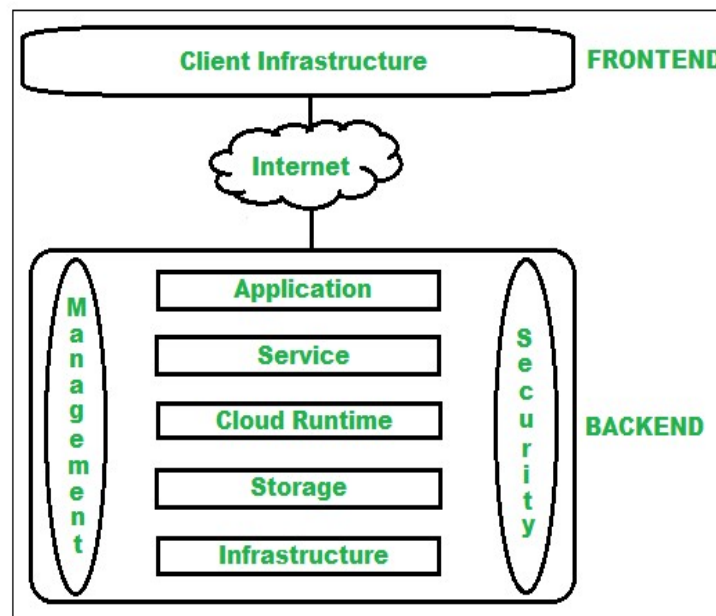
Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.

o **Community Cloud**

5. Observe and note suitable **real-world examples** of each service model and deployment model.
6. Compare **IaaS, PaaS, and SaaS** according to user control, management responsibility, and the resources provided by the cloud provider.
7. Compare the deployment models with respect to **accessibility, ownership, security, resource sharing, and intended users**.
8. Organize the collected observations in suitable **tables and diagrams** representing Cloud Computing architecture, service models, and deployment models.

Stepwise Screenshots with steps:





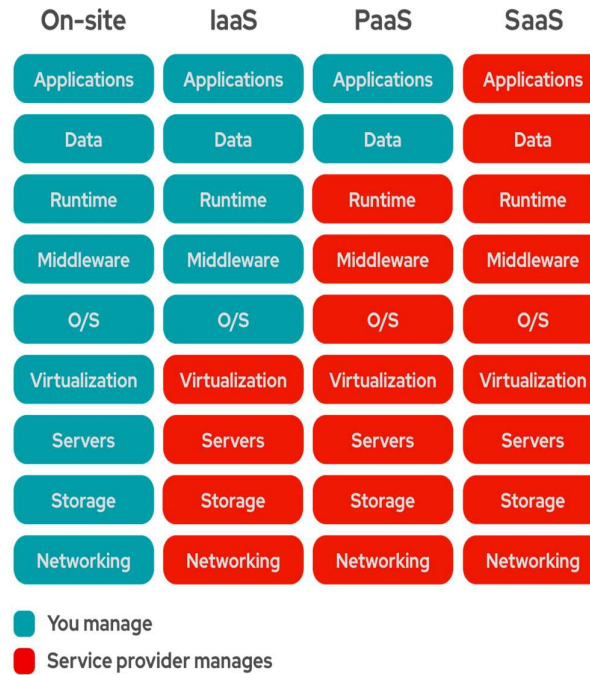
Department of Computer Science & Engineering (IoT)

Vision of the Department

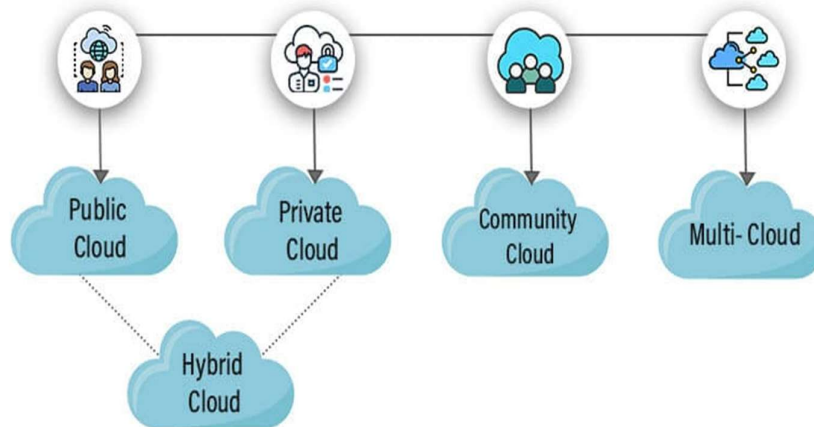
To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.



Cloud Deployment Models



Department of Computer Science & Engineering (IoT)

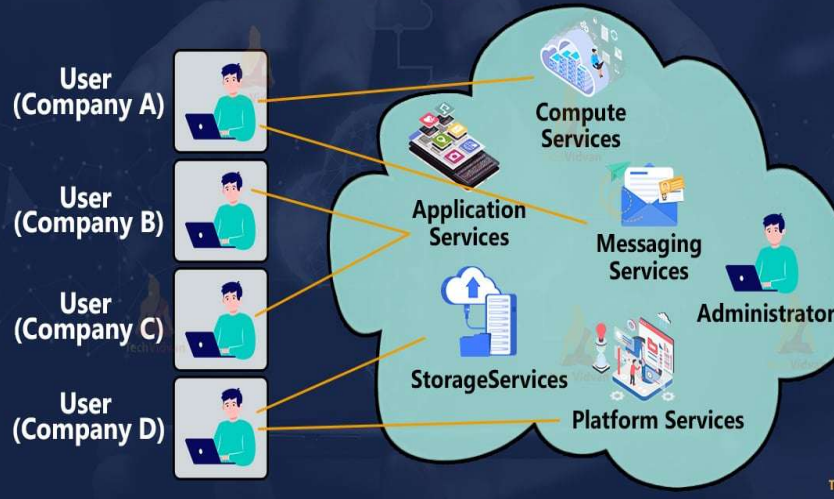
Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

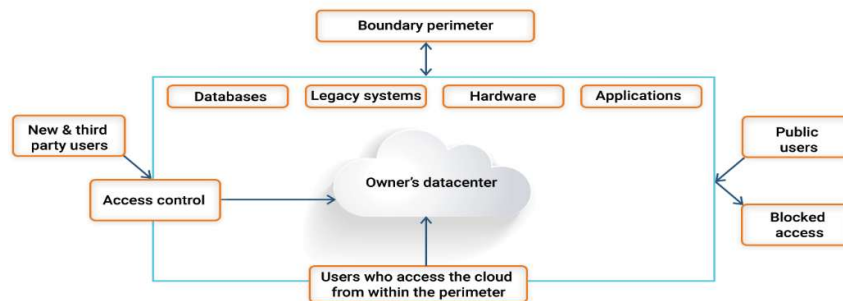
To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.

Cloud Computing- Public Clouds

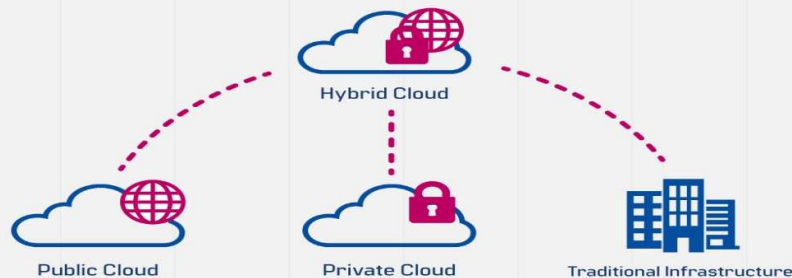


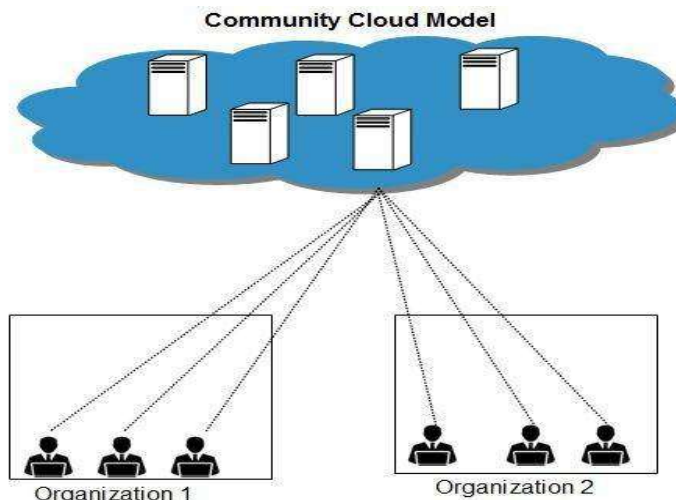
PRIVATE CLOUD

TOOLBOX™



HYBRID CLOUD



**Department of Computer Science & Engineering (IoT)****Vision of the Department***To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.***Mission of the Department***To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.*

	SaaS	PaaS	IaaS
Examples	Netflix Slack Canva	Salesforce Lightning SAP Cloud AWS Lambda	Google Cloud AWS Microsoft Azure
Use cases	Numerous applications across consumer and enterprise.	App development, DevOps, API management	Ecommerce, Disaster recovery, Event processing
Level of Ownership & Control	Low	Medium	High
Integrations Possible	No	Yes	Yes
Easy of use	No Code	Low Code	Low Code/Code Intensive
Need IT Specialists	No	Yes	Yes
Suited for	End Customers or Business Users	Platform Developers	Infrastructure Specialists
Pricing Model	Subscription	Subscription + Professional Services	Subscription + Professional Services



Nagar YuwakShikshanSanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to RashtasantTukadojiMaharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu



Department of Computer Science & Engineering (IoT)

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.

Conclusion	The practical successfully demonstrated the basic concepts of Cloud Computing, its architecture, service models, and deployment models. The roles of the Front End, Back End, and Internet in cloud architecture were studied. The three service models, IaaS, PaaS, and SaaS, were understood based on their control and management responsibilities. The four deployment models, Public, Private, Hybrid, and Community Cloud, were also studied based on their accessibility, ownership, security, and usage. Thus, the practical provided a clear understanding of how cloud services are organized and delivered according to different user and organizational requirements.
Date	03/07/2026